

La'Tahvia (Tahvia) Williams

Data Analyst · Python · SQL · Statistical Modeling · Machine Learning

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SUMMARY

Data Science and Public Policy student at the University of Chicago who ships end to end, from ingestion to deployed application. Led a 3-person team modeling ~300,000 federal sentencing records, built a pipeline over 1.18M payment transactions, and shipped a five-language retrieval app to production. Regression, machine learning and text as data.

EDUCATION

University of Chicago

B.A., Data Science and Public Policy (Double Major) · GPA: 3.3

Chicago, IL

Expected 2028

Coursework: Statistical Models and Regression, Machine Learning, Data Engineering, Big Data, Data Visualization

TECHNICAL SKILLS

Languages: Python, SQL, R, TypeScript, JavaScript

Analysis & ML: pandas, NumPy, scikit-learn, statsmodels, OLS / logistic / LASSO regression, fixed effects, PCA, hypothesis testing

Text & Retrieval: TF-IDF, sentence-transformers, BM25, cross-encoder reranking, Chroma (RAG)

Visualization & Systems: Altair, Plotly, matplotlib, seaborn, D3, GeoPandas, PostgreSQL, Supabase, FastAPI, React, GitHub Actions

EXPERIENCE

Rights Within Reach

Founder & Developer

Chicago, IL

Jun. 2026 – Present

- Built and deployed rightswithinreach.org, a multilingual legal-rights retrieval app indexing 130+ documents across 5 languages; FastAPI/Chroma RAG backend, React/TypeScript frontend on Railway and Vercel.
- Designed the retrieval architecture with metadata pre-filtering, hybrid BM25 search, and cross-encoder reranking, improving retrieval accuracy **89%** over a dense-only baseline.

TectoniQ

Quantitative Research Fellow

Remote

Jun. 2026 – Aug. 2026

- Benchmarked a quantum fidelity kernel against tuned classical baselines, running a model ladder from logistic regression to gradient boosting under equal tuning.
- Overturned three prior over-optimistic findings by rebuilding the evaluation with bootstrap confidence intervals and DeLong testing, establishing no significant edge over gradient boosting.

Spry Digital

Technology Intern

St. Louis, MO

Jun. 2025 – Aug. 2025

- Built REST API integrations and custom GPT workflows automating internal reporting and large-scale data migration.

Armanino LLP · Reinsurance Group of America

Auditing Intern · EPM Intern — 401(k) audits at 99% compliance; training-system rebuild, +30% accuracy

St. Louis, MO

2023 – 2024

PROJECTS

Sentenced Differently — Disparities in Federal Sentencing

Python, statsmodels, pyreadstat

- Led a 3-person team as Project Manager and Tech Lead, analyzing ~300,000 federal sentencing records (FY2020–FY2024) for residual race and gender disparities after legal controls.
- Built a two-stage model separating the incarceration decision (logistic) from sentence length (log-linear OLS) with district and fiscal-year fixed effects; validated with VIF, Cook's distance, robust SEs.

Words and Money at ECOSOC — Text as Data

Python, scikit-learn, TF-IDF, embeddings

- Tested whether UN discourse predicts UN spending across ~1,200 ECOSOC transcripts, 19 agencies and ~22,700 World Bank projects; built features in three tiers (lexicons, TF-IDF, embeddings) with hyperparameters frozen for a clean ablation, cutting OLS out-of-sample RMSE from **9.079** to **0.094**.

Open Payments Analysis — Pharmaceutical Industry Transparency

Python, pandas, Plotly, CMS data

- Analyzed \$157.8M across 1.18M payment transactions to OB/GYN specialists (2017–2020); found a **43%** pandemic-year collapse (\$46.5M → \$26.3M) and a 140% rise in average Reproductive Endocrinology payments.

Forgotten Wars — Quantifying the Attention Hierarchy of Conflict

Python, ACLED, Google Trends

- Correlated 884,216 conflict events across 11 countries against search volume; attention ranged $r = 0.77$ to 0.03 , a 25-fold gap independent of death toll.

The Geography of Rights — 50-State Policy Index

Python, statsmodels, Altair

- Scored all 50 states on a composite Human Rights Policy Index across five domains using PCA weighting; modeled political and demographic predictors via OLS.

PublicPath — Public-Sector Job & Partner Hub

Python, PostgreSQL, GitHub Actions

- Built 20+ Python sync scripts normalizing federal, state, city and nonprofit postings into one Supabase/PostgreSQL schema; automated daily ingestion via GitHub Actions.

LEADERSHIP & ACTIVITIES

- Inter-House Council, President** (prev. VP) & **Resident Assistant:** Residential governance for 4,000+ students; twice-per-quarter funding cycle; support 100+ residents.
- The Chicago Maroon, Data Reporter:** Financial aid analysis benchmarking UChicago against 9 peers using Common Data Sets.
- Greenwood Project, FinTech Scholar:** Selective program placing students into quantitative and financial services roles.